

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	01 October 2026
Team ID	SWTID-2026-2391
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed Script-Controlled ACL solution. The requirements are derived from the demonstrated ServiceNow configuration.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User and Role Management	Create the EEE User and assign custom roles bb1, bb2, bb3 and bb4 for access control.
FR-2	Institution Details Management	Create the custom u_institution_details table with Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description fields.
FR-3	READ Access Control	Allow permitted users to view records through the READ ACL; use the advanced server-side script to allow administrators and users with bb1, while denying others.
FR-4	CREATE Access Control	Allow record creation only when the user has the required bb2 role.
FR-5	WRITE Access Control	Allow record modification only when the user has the required bb3 role.
FR-6	DELETE Access Control	Allow record deletion only when the user has the required bb4 role.
FR-7	Branch-based Record Access	Support ECE, EEE and CSE branch values and enforce the demonstrated EEE-related record visibility scenario through ACL configuration.
FR-8	Administrator Access	Provide full administrative access to records and permit ACL configuration with appropriate security privileges.
FR-9	Access Verification	Allow verification of ACL behavior through user impersonation and observation of Student Records lists/forms and CRUD capabilities.

Non-functional Requirements:

Following are the non-functional requirements applicable to the demonstrated ServiceNow ACL solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The access result should be clear through the ServiceNow Student Records list and record forms. Users should see only the actions permitted by their assigned roles.

NFR-2	Security	Record access must be controlled by ServiceNow ACLs, roles and the advanced READ script. Unauthorized operations must be denied. ACL configuration requires appropriate administrative/security privileges.
NFR-3	Reliability	ACL behavior should remain consistent with the configured role rules. Changes should be verified by impersonating relevant users and checking the expected access result.
NFR-4	Performance	Access decisions should be evaluated through the ServiceNow ACL engine and the configured server-side READ script without introducing unnecessary application-level processing.
NFR-5	Availability	The solution depends on the availability of the ServiceNow instance and its authentication, authorization and table services.
NFR-6	Scalability	The ACL approach can be extended to additional branches, departments, data conditions, roles and custom tables as proposed in the project documentation.